Voice Scribe — EDA Findings & Next Steps

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Summary

The EDA checked what speech-to-text models will face in the English test set and whether that changes how we score them. It does, in four ways:

1. WER must be pooled over all words, not averaged per clip: 18% of Svarah clips are under 1 s, almost all a single word.
2. Doctor vs patient speech and each Eka clip type need separate scores.
3. Drug names need their own accuracy score: 60% of drug mentions contain a word rare in general English.
4. Numbers and units need a scoring rule (v2, proposed below): 20% of Eka clips have a dosage, written inconsistently.

The noisy versions match their labelled levels closely enough to compare models. Next: evaluate the candidate models on English, then Hindi on DISPLACE-M once access arrives.

All decisions below were made from the dev split only; the test split is kept for final results.

1. Size and balance

Clip lengths range from one-word commands under a second to 9-minute consultations, so per-clip averaging would distort the score. Eka speaker counts exclude its 110 conversation clips, whose speakers are not identified; Svarah counts are speaker profiles (115 profiles for about 117 speakers).

Dataset	Dev	Test	Typical length
PriMock57 consultations	11 consultations, 1.7 h	46 consultations, 7.0 h	~9 min
Eka Medical ASR (en)	735 clips, 11 speakers, 1.6 h	2,884 clips, 45 speakers, 6.8 h	2.5–6 s (single terms), 11–20 s (sentences), 10th–90th percentile

Dataset	Dev	Test	Typical length
Svarah	1,351 clips, 19 speakers, 2.2 h	5,305 clips, 96 speakers, 7.4 h	median 4–5 s; 18% under 1 s

Implication: compute WER over all words pooled together. A one-word clip with one error is 100% WER, and averaging clips would let those dominate.

2. PriMock57: how the consultations behave

Doctors dominate the conversation and patients speak in short bursts, so the two need separate scores.

Measure	Doctor	Patient
Share of annotated speech time	59% on average (37–77% per consultation)	41%
Speaking rate (per minute of annotated utterance)	174 words/min	151 words/min
Median utterance	11 words, 4.1 s	6 words, 2.5 s
Utterances with an unintelligible stretch	19%	15%
Utterances with an uncertain stretch	15%	15%

Doctor and patient utterances overlap for 4.8% of the consultation on average, up to 13.9% in one consultation.

Implications:

- Report WER separately for doctor and patient speech.
- The overlap makes even the clean set a mild test of handling two speakers.
- Keep flagged utterances in the main score, plus a strict score that excludes them.

3. Eka: numbers, units and medical terms

Half of Eka's transcripts contain a number and a fifth contain a dosage, written in several styles, so formatting alone would cost models points without a rule.

Pattern	Dev clips	All clips
Any digit	49.5%	47.6%
Number + unit ("40Mg", "100mg", "12.5 Mg")	20.1%	20.5%
Drug name containing digits ("Dolo 650"), from entity tags	33.2%	32.8%
Number written as a word ("one", "five"), approximate	4.9%	3.9%
Once / twice / thrice	0.3%	0.7%
Dosing abbreviation (OD, BD, TDS, SOS)	0.5%	0.6%
Ordinal ("3rd", "22nd")	0.1%	0.2%

Medical terms tagged in Eka: 5,615 clinical findings, 2,504 drugs (1,891 distinct), 1,145 other medical terms, 1,021 diagnostics, 376 advices. Common drugs include Dolo 650, amlodipine, metformin and thyroxine. Some entries tagged as drugs are manufacturer names ("Intas Pharmaceuticals Limited").

Implications:

- A scoring rule for numbers and units is needed (rule v2, section 7).
- Report WER per clip type: single terms, dictated sentences, conversation.

4. Svarah: who is speaking

Svarah covers 115 speaker profiles (about 117 speakers) with 19 first languages from 17 states in the metadata (the dataset card says 19), enough to report accuracy by accent group, though some groups are small.

- **Largest groups:** Tamil (10 speakers), Kannada (9), Telugu (8).
- **Smallest groups:** Sindhi (2), Kashmiri (4); four languages have 5 each.
- **Gender (per profile):** 62 female, 53 male. **Age:** 18–30: 37, 30–45: 41, 45–60: 27, 60+: 10.
- **Dev split:** covers 15 of the 19 first languages.

- **Short clips:** 18% are under 1 s, and 93% of those are a single word ("Jump", "Play", "Skip").

Implication: treat accent-group results for groups of 2–5 speakers as indicative only.

5. Noisy versions: do the levels match the labels?

Every dev noisy file was re-measured by subtracting the clean consultation and comparing what remains with the consultation's speech. Every condition measures on target, within 0.2 dB. Interference is measured over the background talker's own speech (as the levels were set) and over the whole file for noise.

Condition	Labelled level (dB below consultation)	Measured mean	Measured range
Clinic noise	20 / 10 / 5	20.0 / 10.0 / 5.0	exact
Indian background talker	10 / 5 / 0	9.9 / 5.0 / 0.0	within 0.2
British background talker	10 / 5 / 0	10.0 / 5.0 / 0.0	within 0.2
Busy OPD (talker 5 + noise 10 combined; expected 3.8)	–	4.1	3.8–4.4

Spectrograms of the same 20 s under each condition show background speech and noise filling the pauses between the consultation's speech, as intended.

Implication: the noisy set's levels can be quoted as labelled. Busy OPD measures 0.3 dB above 3.8 because its talker is measured over the whole file, pauses included.

6. Vocabulary gap: words models will likely miss

Drug names are the hardest vocabulary in the set: 60% of drug mentions contain a word that is rare in general English text (fewer than ~3 uses per 100 million words, by `wordfreq`).

Term type	Mentions	Containing a rare word
Drugs	2,504	59.6%

Term type	Mentions	Containing a rare word
Diagnostics	1,021	10.9%
Advices	376	10.6%
Clinical findings	5,615	6.5%
Other medical terms	1,145	3.5%
PriMock57, all words	–	0.17%

Examples of rare drug terms: Zincovit, Pantop DSR, Glycomet Trio, Azoran, Volini, Embramine, Etoshift.

Implications:

- Score drug-name accuracy separately from WER.
- The datasets test different things: PriMock57 tests noise and conversation, Eka tests medical vocabulary.

7. Scoring decisions

The EDA fixes how every model will be scored; rule v2 is proposed and awaits approval.

How results are reported:

Dataset	Reported separately for
PriMock57	doctor vs patient; each of the 11 conditions; main and strict (flagged utterances excluded)
Eka	single terms, dictated sentences, conversation; drug-name accuracy
Svarah	overall; by first-language group

WER is always pooled over all words, never averaged per clip.

Scoring rule v2 (proposed), applied identically to the reference and to each model's output:

1. Split numbers from units: "40Mg" → "40 mg".
2. One form per unit: milligram → mg, millilitre → ml, microgram → mcg, gram → g.

3. Number words to digits: "five hundred" → "500".
4. Leave "once / twice / thrice" and dosing abbreviations unchanged (under 1% of clips).

Open: whether "2 times a day" and "twice a day" should count as equal; to be decided once real model outputs show how often it matters.

8. Models to evaluate

Nine candidates cover open models we can run ourselves and cloud services, chosen because each either leads a relevant published benchmark or targets Indian speech. Exact versions are fixed at test time to the latest available.

Model	Type	Why it's on the list	Published result to compare against
Whisper large-v3 (OpenAI)	Open weights	Standard multilingual baseline in every comparison	15.7% WER on Eka (en); ~98.6% on real Kannada clinical audio
Parrotlet-a-en-5b (Eka Care)	Open weights	Built for Indian medical English	10.9% WER on Eka (en), best published
IndicConformer 600M (AI4Bharat)	Open weights	Indian-language ASR; DISPLACE-M baseline	25.6% WER on DISPLACE-M zero-shot
IndicWhisper (AI4Bharat)	Open weights	Whisper adapted to Indian languages	tested in ASR Under the Stethoscope
Qwen3-ASR 1.7B	Open weights	Base of the best DISPLACE-M challenge system	18.2% WER on DISPLACE-M after fine-tuning
Gemini 2.5 Pro / Flash (Google)	Cloud API	Best on real Indian hospital audio	14.2% English, 18.5% Hindi, 35.0% Kannada; 14.8% on Eka (Flash)
Google Chirp 3	Cloud API	Managed multilingual service	tested in ASR Under the Stethoscope
Amazon Transcribe	Cloud API	Managed service with Indian-language options	18.3% WER on Eka (en)

Model	Type	Why it's on the list	Published result to compare against
Sarvam (Saarika / Saaras)	Cloud API	Indian-language specialist	24.6% WER on DISPLACE-M; high insertion (hallucination) rates reported

What each model is measured on: WER (pooled), CER, drug-name accuracy, background-speech leakage (words from the known background conversation), speed and cost per hour of audio.

Order: run the open models first on the dev split, then the cloud APIs, then the shortlist on test.

9. DISPLACE-M: what it contains and how we'll use it

DISPLACE-M is the only dataset found with real, noisy Indian healthcare conversations, so it moves the project from simulated noise to real field audio. Registration for tracks 1 (diarization), 2 (ASR) and 4 (summarisation) is in progress.

What the data contains (from the challenge papers):

Aspect	Detail
Speakers and setting	Community health workers (ASHAs) and patients in rural and semi-urban India: health centres, Anganwadi centres, schools, homes; indoors and outdoors
Language	Hindi with code-switching to English and dialects (Haryanvi, Bhojpuri, Magahi); English words written in Devanagari
Size	~35 h for development and evaluation, 260 speakers
Tracks 1–2 (diarization, ASR)	25 h development (two sets of 15 h and 10 h) + 10 h blind evaluation
Tracks 3–4 (topics, summaries)	~15 h: 10 h development + 5 h evaluation

Aspect	Detail
Recording	Phone microphone at a distance, 16 kHz; variable room acoustics and field noise
Speech mix	84% one speaker, 11% silence, 4% overlapping speech
Annotations	Speaker turns with timestamps; verbatim transcripts (ASR draft corrected by humans); topic labels; summaries written by a clinician
Licence	MIT; redistribution needs organisers' consent; a 2–4 page system report is expected from participants

How we will use it:

1. **Hindi model selection:** run the same candidates on real noisy clinical Hindi, scoring WER and CER. Because the blind evaluation labels are hidden, we split the 25 h development data into our own dev and test, by speaker.
2. **Reality check for the simulation:** compare how models rank on DISPLACE-M against their ranking on noisy PriMock57. If the rankings agree, the simulated set is a valid proxy.
3. **Speaker attribution:** use the speaker-turn labels to evaluate diarization, the step that separates the consultation from other voices.
4. **Clinical notes later:** the clinician-written summaries give a reference for testing note generation from transcripts.

Scoring adjustments it needs: English words in Devanagari mean "blood test" may appear as ब्लड टेस्ट, so outputs must be normalised to one script, with CER reported alongside WER.

Published results to compare against: IndicConformer zero-shot 25.6% WER, fine-tuned 19.0%; Whisper large-v3 fine-tuned 23.6%; Gemini 2.5 Pro 19.6%; Sarvam Saaras v3 24.6%; best challenge system (fine-tuned Qwen3-ASR) 18.2%.

10. Next steps

English (now):

- [ ] Approve scoring rule v2 and rebuild the reference transcripts with it

- [ ] Build the scoring script: pooled WER, CER, drug-name accuracy, leakage, speed, cost
- [ ] Decide where models run (GPU or cloud APIs) and set up access
- [ ] Run open models on dev, then cloud APIs on dev
- [ ] Shortlist, then run the shortlist on test and write up results

Hindi and Kannada (when access arrives):

- [ ] Complete DISPLACE-M registration with the signed terms
- [ ] Request DISPLACE 2024 access and accept IndicVoices terms
- [ ] Split DISPLACE-M development data into our own dev and test by speaker
- [ ] Add script normalisation (Devanagari vs Latin) to the scoring
- [ ] Run the candidates and compare rankings with the simulated English results

Real-world check:

- [ ] Plan 20–30 staged OPD recordings: consent, phone placement, script outline

Sources

- PriMock57 — Papadopoulos Korfiatis et al., ACL 2022
- Eka Medical ASR Evaluation Dataset — dataset card with model results
- Svarah — Javed et al., Interspeech 2023
- MS-SNSD noise dataset — Reddy et al., 2019
- Benchmarking Speech Systems for Frontline Health Conversations: The DISPLACE-M Challenge
- DASHA: Diarization-Aware Speech-LLM ASR for Code-Switched Indic Doctor-Patient Conversations
- ASR Under the Stethoscope: Evaluating Biases in Clinical Speech Recognition across Indian Languages
- wordfreq — word-frequency data used for the vocabulary-gap check